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Experiment No. 1: Comparative Study of Low-Code Development versus Traditional Application Development Approaches.

Aim

To examine the differences between conventional software development and low-code platforms by comparing their workflow, productivity, flexibility, cost, maintenance, and business suitability.

Introduction to Conventional Software Development

Conventional software development involves creating applications by writing source code using programming languages and frameworks. Every phase, including requirement analysis, design, coding, testing, deployment, and maintenance, is handled by software professionals. This approach offers excellent scalability and customization but requires greater development effort.

Key Characteristics:

- Code-centric development
- Suitable for enterprise systems
- High flexibility
- Longer development cycle
- Requires experienced developers

Introduction to Low-Code Development

Low-code development uses visual modelling, reusable components, and workflow automation to develop applications quickly with minimal programming. It enables organizations to deliver applications faster while reducing manual coding.

Key Characteristics:

- Visual interface
- Reusable components
- Rapid development
- Easy integration
- Faster deployment

Workflow of Conventional Development

Requirements

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Planning

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System Design

↓
Coding
↓
Testing
↓
Deployment
↓
Maintenance

Workflow of Low-Code Development

Business Requirements
↓
Visual App Design
↓
Configure Logic
↓
Connect Data Sources
↓
Testing
↓
Publish Application
↓
Continuous Improvement

Comparative Analysis

Aspect	Conventional Development	Low-Code Development
Development Time	Long	Short
Coding	Complete manual coding	Minimal coding
Budget	Higher	Lower
Flexibility	Excellent	Moderate
Maintenance	More effort	Simpler
Users	Software engineers	Developers & business users
Testing	Mostly manual	Tool-assisted
Ideal For	Complex enterprise systems	Rapid business applications

Result

Both approaches have unique strengths. Conventional development is the preferred option for highly customized and mission-critical applications, whereas low-code platforms are better suited for rapid application delivery, business process automation, and projects with shorter timelines.